

3(a)	<i>any four from:</i> 1 <i>ref. to</i> geographical isolation; 2 no, gene flow / breeding, between populations (on the different islands) ; 3 different, selection pressures / environmental conditions (on the different islands) ; 4 different mutations occur (on the different islands) ; 5 some mutations make individuals better adapted ; 6 those individuals, survive / reproduce ; 7 pass on advantageous alleles ; 8 <i>ref. to</i> many generations ; 9 reproductive isolation ; 10 allopatric speciation ;	4
3(b)	<i>any two from:</i> 1 bind to complementary base sequences in mtDNA ; 2 (so) only amplify specific mtDNA sections / AW ; 3 mtDNA section, differences / similarities, used to assess how closely related the subspecies ;	2

Question	Answer	Marks
3(c)	<i>any one pair from:</i> 1 large quantity in the cell ; 2 (so) easier to, extract / amplify, DNA for testing ; <i>or</i> 3 small genome size ; 4 (so) easier to locate specific section of DNA to test ; <i>or</i> 5 mtDNA is, a single copy of DNA / not paired alleles / haploid ; 6 (so) only mutation causes it to change ; <i>or</i> 7 inherited maternally ; 8 (so) all mtDNA sections are shared between all members (of maternal) family ; <i>or</i> 9 mutation rate is higher / no enzymes to repair mutations / faster molecular clock ; 10 (so) more choice of suitable sections of mtDNA to test / more accurate time estimate ;	2
3(d)	<i>any two from:</i> 1 habitat loss ; 2 more competition for, space / mates ; 3 reduction in prey ; 4 poaching / hunting / poisoning (by humans) ; 5 disease ;	2
3(e)	<i>any two from:</i> 1 morphological / physiological / behavioural / biochemical, similarity to mainland tigers / <i>Panthera tigris</i> ; 2 genetically similar to, mainland tigers / <i>Panthera tigris</i> ; 3 may be able to breed with, mainland tigers / <i>Panthera tigris</i>, to produce fertile offspring ;	2